



Qualification Pack



Retrofitting Technician (Construction Sector)

QP Code: CBWE/CON/Q0101

Version: 1.0

NSQF Level: 4

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Qualification Pack

Contents

CBWE/CON/Q0101: Retrofitting Technician (Construction Sector)	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
CBWE/CON/N0101: Assess Building Condition and Identify Repair & Strengthening Requirements	5
CBWE/CON/N0102: Conduct Structural Repairs (Masonry, Plastering, Waterproofing, etc.)	9
CBWE/CON/N0103: Perform Crack and Joint Repairs (Epoxy, Polyurethane Injection)	13
CBWE/CON/N0104: Apply Structural Re-Strengthening Techniques (Steel, Concrete, FRP)	17
CBWE/CON/N0105: Implement Seismic Retrofitting Solutions	21
CBWE/CON/N0106: Execute Quality Control and Safety Standards in Retrofitting Work	25
DGT/VSQ/N0102: Employability Skills (60 Hours)	29
Assessment Guidelines and Weightage	36
<i>Assessment Guidelines</i>	36
<i>Assessment Weightage</i>	37
Acronyms	38
Glossary	39



Qualification Pack

CBWE/CON/Q0101: Retrofitting Technician (Construction Sector)

Brief Job Description

The Retrofitting Technician at NSQF Level 4 is skilled in performing complex retrofitting operations to repair and strengthen existing structures. This qualification is tailored to individuals involved in building repair and re-strengthening, utilizing techniques such as structural repairs, seismic retrofitting, crack injections, and strengthening with steel, concrete, and other materials. The role requires in-depth technical knowledge, practical expertise, and adherence to safety standards to ensure that buildings are restored to a condition that enhances their durability and load-bearing capacity.

Personal Attributes

The individual should be physically fit to work for sustained durations in varying site conditions and environments. The person should be able to interpret basic structural drawings, follow technical instructions, and work in coordination with structural engineers and other team members. The individual should have a safety-conscious attitude, sound manual dexterity, and adequate verbal communication skills to report observations accurately.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CBWE/CON/N0101: Assess Building Condition and Identify Repair & Strengthening Requirements](#)
2. [CBWE/CON/N0102: Conduct Structural Repairs \(Masonry, Plastering, Waterproofing, etc.\)](#)
3. [CBWE/CON/N0103: Perform Crack and Joint Repairs \(Epoxy, Polyurethane Injection\)](#)
4. [CBWE/CON/N0104: Apply Structural Re-Strengthening Techniques \(Steel, Concrete, FRP\)](#)
5. [CBWE/CON/N0105: Implement Seismic Retrofitting Solutions](#)
6. [CBWE/CON/N0106: Execute Quality Control and Safety Standards in Retrofitting Work](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Construction
Sub-Sector	Building Construction
Occupation	Building Retrofitting



Qualification Pack

Country	India
NSQF Level	4
Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7119.9900
Minimum Educational Qualification & Experience	10th grade pass with 3 Years of experience in relevant field OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	1.0
Reference code on NQR	QG-04-CO-04329-2025-V1
NQR Version	1

Remarks:

As per the Government Guidelines



Qualification Pack

CBWE/CON/N0101: Assess Building Condition and Identify Repair & Strengthening Requirements

Description

This unit describes the skills and knowledge required to inspect and evaluate building structural elements, determine root causes of structural weaknesses, review relevant drawings, and select appropriate repair and strengthening techniques in alignment with national standards.

Scope

The scope covers the following :

- Inspect and evaluate structural condition of buildings
- Review documentation and select repair and strengthening strategies

Elements and Performance Criteria

Inspect and evaluate structural condition of buildings

To be competent, the user/individual on the job must be able to:

- PC1.** inspect and evaluate building structural elements (e.g., beams, columns, foundations) to identify visible signs of damage such as cracks, corrosion, and decay
- PC2.** determine the root causes of structural weaknesses, such as water leakage, settlement, or wear and tear

Review documentation and select repair and strengthening strategies

To be competent, the user/individual on the job must be able to:

- PC3.** review architectural and structural drawings to understand original design and current condition
- PC4.** select appropriate repair and strengthening techniques based on the severity and type of damage
- PC5.** ensure that the assessment aligns with relevant national standards and codes for building repair and strengthening

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** load-bearing functions and structural behaviour of beams, columns, slabs, foundations, and load-bearing walls
- KU2.** effect of deterioration or damage in structural elements on the stability and integrity of the overall building
- KU3.** common signs and causes of structural damage, including cracks, spalling, corrosion, settlement, water ingress, and material fatigue
- KU4.** types of architectural and structural drawings including plan drawings, elevation drawings, and structural detail drawings



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- KU5.** different methods of structural condition assessment including visual inspection, non-destructive testing (NDT), and field testing techniques
- KU6.** commonly used repair and strengthening techniques for different types and severity of structural damage
- KU7.** relevant national standards and IS codes applicable to building inspection, repair, and retrofitting
- KU8.** criteria for classifying structural damage severity, including damage rating indicators and conditions requiring immediate intervention or engineer referral
- KU9.** visible effects of structural element deterioration on overall building performance and stability
- KU10.** types of temporary stabilisation measures including propping, shoring, strutting, and needling used to support unsafe structures or elements
- KU11.** conditions requiring access restriction or evacuation, and the reporting procedure for unsafe structural conditions to supervisor

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and follow work instructions, drawings, and site documents
- GS2.** write accurate records and reports of site observations and findings
- GS3.** communicate findings clearly to supervisors and team members
- GS4.** listen to and follow instructions given by supervisors
- GS5.** identify safety hazards at the worksite and report to supervisors
- GS6.** plan and prioritise work activities based on urgency and agreed schedule
- GS7.** evaluate task complexity and seek guidance from supervisors when required
- GS8.** identify any violation of safety norms or unsafe conditions and report immediately
- GS9.** apply systematic observation skills to identify and record visible defects accurately
- GS10.** take immediate action to restrict access and report unsafe structural conditions to the supervisor
- GS11.** arrange or apply temporary stabilisation measures to support an unsafe structural element



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Inspect and evaluate structural condition of buildings</i>	12	24	-	4
PC1. inspect and evaluate building structural elements (e.g., beams, columns, foundations) to identify visible signs of damage such as cracks, corrosion, and decay	-	-	-	-
PC2. determine the root causes of structural weaknesses, such as water leakage, settlement, or wear and tear	-	-	-	-
<i>Review documentation and select repair and strengthening strategies</i>	18	36	-	6
PC3. review architectural and structural drawings to understand original design and current condition	-	-	-	-
PC4. select appropriate repair and strengthening techniques based on the severity and type of damage	-	-	-	-
PC5. ensure that the assessment aligns with relevant national standards and codes for building repair and strengthening	-	-	-	-
NOS Total	30	60	-	10



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National Occupational Standards (NOS) Parameters

NOS Code	CBWE/CON/N0101
NOS Name	Assess Building Condition and Identify Repair & Strengthening Requirements
Sector	Construction
Sub-Sector	
Occupation	Building Retrofitting
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

CBWE/CON/N0102: Conduct Structural Repairs (Masonry, Plastering, Waterproofing, etc.)

Description

This unit describes the skills and knowledge required to prepare damaged surfaces, apply appropriate repair materials and coatings, follow specific repair methodologies based on material type, and monitor the curing and setting process to ensure structural repair work is completed to standard.

Scope

The scope covers the following :

- Prepare surfaces for structural repair work
- Apply repair materials and monitor curing process

Elements and Performance Criteria

Prepare surfaces for structural repair work

To be competent, the user/individual on the job must be able to:

- PC1.** prepare the surface of damaged walls, ceilings, or floors by cleaning and removing loose material

Apply repair materials and monitor curing process

To be competent, the user/individual on the job must be able to:

- PC2.** apply repair materials such as mortar, plaster, and waterproof coatings to restore the structural integrity of the building
- PC3.** follow specific repair methodologies based on material types (e.g., plaster repairs, brickwork repairs, waterproofing applications)
- PC4.** monitor the curing and setting process to ensure the repair work is completed to standard

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the types and properties of repair materials including cement mortar, polymer-modified mortars, plaster compounds, waterproofing membranes, and surface coatings
- KU2.** surface preparation techniques for different substrates including concrete, masonry, and plastered surfaces, such as hacking, grinding, and wire brushing
- KU3.** standard methodologies for masonry repair including tuck pointing, repointing, and brick or block replacement procedures
- KU4.** types of waterproofing systems including crystalline waterproofing, cementitious coatings, bituminous membranes, and injection grouting
- KU5.** plastering techniques for repair work including the application of scratch coat, brown coat, and finish coat
- KU6.** curing requirements, methods, and duration for different categories of repair materials



Qualification Pack

- KU7.** relevant IS codes and standards for structural repair materials, mix designs, and repair methodologies
- KU8.** factors affecting bond strength of repair materials including surface moisture, ambient temperature, and carbonation depth
- KU9.** selection criteria for appropriate repair materials based on type, location, and extent of structural damage

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and follow work instructions, method statements, and safety guidelines
- GS2.** write accurate records of materials used, work done, and observations
- GS3.** communicate work progress and issues to supervisors and team members
- GS4.** listen to and follow instructions from supervisors
- GS5.** identify safety hazards and use personal protective equipment as required
- GS6.** plan and organise materials and tools before commencing work
- GS7.** complete work as per agreed quality standards and time schedule
- GS8.** work cooperatively with other team members and trades on site
- GS9.** follow the correct sequence of repair activities as instructed and per method statements
- GS10.** check the quality of repair work at each stage before proceeding to the next



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare surfaces for structural repair work</i>	8	15	-	2
PC1. prepare the surface of damaged walls, ceilings, or floors by cleaning and removing loose material	-	-	-	-
<i>Apply repair materials and monitor curing process</i>	22	45	-	8
PC2. apply repair materials such as mortar, plaster, and waterproof coatings to restore the structural integrity of the building	-	-	-	-
PC3. follow specific repair methodologies based on material types (e.g., plaster repairs, brickwork repairs, waterproofing applications)	-	-	-	-
PC4. monitor the curing and setting process to ensure the repair work is completed to standard	-	-	-	-
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CBWE/CON/N0102
NOS Name	Conduct Structural Repairs (Masonry, Plastering, Waterproofing, etc.)
Sector	Construction
Sub-Sector	
Occupation	Building Retrofitting
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

CBWE/CON/N0103: Perform Crack and Joint Repairs (Epoxy, Polyurethane Injection)

Description

This unit describes the skills and knowledge required to identify and assess cracks and joints in structures, select and prepare appropriate injection materials, inject materials to restore structural integrity, and monitor the curing process to ensure effective sealing and bonding.

Scope

The scope covers the following :

- Identify and assess cracks and joints requiring repair
- Execute crack injection repair and monitor curing

Elements and Performance Criteria

Identify and assess cracks and joints requiring repair

To be competent, the user/individual on the job must be able to:

PC1. identify cracks and joints that need repair and assess their depth, width, and severity

PC2. select and prepare appropriate materials (epoxy, polyurethane, etc.) for crack injection

Execute crack injection repair and monitor curing

To be competent, the user/individual on the job must be able to:

PC3. inject the chosen material into the cracks to restore the structural integrity of the building

PC4. monitor the curing process to ensure effective bonding and sealing of the cracks

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** types of cracks in concrete and masonry structures, including shrinkage cracks, structural cracks, and settlement cracks, and their typical causes
- KU2.** crack severity assessment criteria including crack width classification, depth measurement using gauges and probes, and activity monitoring using tell-tales
- KU3.** properties and structural applications of epoxy resin systems for crack repair and structural bonding
- KU4.** properties and applications of polyurethane injection materials for crack sealing, water stop, and flexible joint filling
- KU5.** equipment and tools used for crack injection including low-pressure injection ports, packers, high-pressure injection equipment, and hand-operated grout pumps
- KU6.** stages of the crack injection process including port placement, injection sequence, and completion indicators
- KU7.** curing requirements and post-injection inspection and monitoring procedures



Qualification Pack

- KU8.** safe handling and storage requirements for chemical injection materials including epoxy resins and polyurethane systems
- KU9.** Material Safety Data Sheet (MSDS) interpretation and chemical waste segregation and disposal requirements under applicable environmental regulations
- KU10.** relevant IS codes and standards for crack repair and chemical injection techniques in concrete structures

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and follow product instructions, data sheets, and work method statements
- GS2.** write accurate records of observed defects, materials used, and work completed
- GS3.** communicate work progress and observations to supervisors and team members
- GS4.** listen to and follow instructions from supervisors
- GS5.** handle hazardous materials safely and use appropriate personal protective equipment
- GS6.** pay attention to detail and check work at each stage for accuracy
- GS7.** seek assistance from supervisors when encountering unfamiliar situations or materials
- GS8.** keep tools and the work area clean and organised after completing tasks
- GS9.** use measuring tools and gauges accurately to assess crack dimensions and severity
- GS10.** follow specified mixing ratios and injection sequences as per product instructions



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify and assess cracks and joints requiring repair</i>	15	30	-	5
PC1. identify cracks and joints that need repair and assess their depth, width, and severity	-	-	-	-
PC2. select and prepare appropriate materials (epoxy, polyurethane, etc.) for crack injection	-	-	-	-
<i>Execute crack injection repair and monitor curing</i>	15	30	-	5
PC3. inject the chosen material into the cracks to restore the structural integrity of the building	-	-	-	-
PC4. monitor the curing process to ensure effective bonding and sealing of the cracks	-	-	-	-
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CBWE/CON/N0103
NOS Name	Perform Crack and Joint Repairs (Epoxy, Polyurethane Injection)
Sector	Construction
Sub-Sector	
Occupation	Building Retrofitting
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

CBWE/CON/N0104: Apply Structural Re-Strengthening Techniques (Steel, Concrete, FRP)

Description

This unit describes the skills and knowledge required to evaluate the need for structural strengthening, prepare surfaces for the application of strengthening materials, install and secure reinforcement materials including steel plates and fiber-reinforced polymers (FRP), and ensure that re-strengthening work integrates seamlessly with existing structures and complies with engineering standards.

Scope

The scope covers the following :

- Evaluate strengthening requirements and prepare surfaces
- Install reinforcement materials and ensure compliance

Elements and Performance Criteria

Evaluate strengthening requirements and prepare surfaces

To be competent, the user/individual on the job must be able to:

- PC1.** evaluate the need for structural strengthening based on load-bearing requirements or damage analysis
- PC2.** prepare the surfaces for the application of strengthening materials (e.g., steel plates, carbon fiber wraps)

Install reinforcement materials and ensure compliance

To be competent, the user/individual on the job must be able to:

- PC3.** install and secure reinforcement materials, including steel plates or fiber-reinforced polymers (FRP), as per engineering specifications
- PC4.** ensure the re-strengthening work is integrated seamlessly with existing structures and complies with building codes

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** structural strengthening methods including externally bonded steel plates, externally bonded FRP systems, and reinforced concrete jacketing
- KU2.** properties and characteristics of structural strengthening materials including structural steel, CFRP, GFRP, and AFRP
- KU3.** surface preparation requirements for bonding strengthening materials including grinding, sandblasting, and priming
- KU4.** process of steel plate bonding including adhesive selection, application sequence, and mechanical anchoring



Qualification Pack

- KU5.** FRP strengthening installation including wet lay-up and pre-cured laminate techniques and quality requirements for each
- KU6.** quality control inspections required during and after installation of strengthening systems, including bond testing
- KU7.** relevant IS codes, CPWD guidelines, and engineering standards for structural strengthening and rehabilitation of buildings
- KU8.** purpose and application of strengthening techniques including steel plate bonding, FRP wrapping, and concrete jacketing
- KU9.** information contained in engineering specifications and method statements relevant to the installation of strengthening systems
- KU10.** health, safety, and environmental requirements when working with FRP materials, chemical adhesives, and heavy structural components

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and follow engineering drawings, specifications, and work instructions accurately
- GS2.** write accurate records of materials used, installation activities, and observations
- GS3.** communicate work progress and deviations to supervisors and team members promptly
- GS4.** listen to and follow instructions from supervisors and engineers on site
- GS5.** identify safety hazards when working with heavy materials and tools
- GS6.** plan and sequence work activities to ensure quality and minimise disruption
- GS7.** work cooperatively with other trades and team members on site
- GS8.** complete work to required quality standards within the agreed time frame
- GS9.** follow engineering specifications and method statements precisely without deviation
- GS10.** verify fit and alignment of strengthening materials before final fixing



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Evaluate strengthening requirements and prepare surfaces</i>	15	30	-	5
PC1. evaluate the need for structural strengthening based on load-bearing requirements or damage analysis	-	-	-	-
PC2. prepare the surfaces for the application of strengthening materials (e.g., steel plates, carbon fiber wraps)	-	-	-	-
<i>Install reinforcement materials and ensure compliance</i>	15	30	-	5
PC3. install and secure reinforcement materials, including steel plates or fiber-reinforced polymers (FRP), as per engineering specifications	-	-	-	-
PC4. ensure the re-strengthening work is integrated seamlessly with existing structures and complies with building codes	-	-	-	-
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CBWE/CON/N0104
NOS Name	Apply Structural Re-Strengthening Techniques (Steel, Concrete, FRP)
Sector	Construction
Sub-Sector	
Occupation	Building Retrofitting
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

CBWE/CON/N0105: Implement Seismic Retrofitting Solutions

Description

This unit describes the skills and knowledge required to analyse existing buildings for seismic vulnerability, install seismic retrofitting measures including shear walls, base isolation systems, and bracing systems, ensure compliance with seismic safety codes and standards, and test and verify the effectiveness of installed seismic retrofitting measures.

Scope

The scope covers the following :

- Analyse seismic vulnerability of existing buildings
- Install, ensure compliance and verify seismic retrofitting measures

Elements and Performance Criteria

Analyse seismic vulnerability of existing buildings

To be competent, the user/individual on the job must be able to:

PC1. analyze existing buildings for seismic vulnerability based on building age, design, and location

Install, ensure compliance and verify seismic retrofitting measures

To be competent, the user/individual on the job must be able to:

PC2. install seismic retrofitting measures such as shear walls, base isolation systems, or bracing systems as required by the building's seismic evaluation

PC3. ensure that retrofitting techniques comply with seismic safety codes and standards

PC4. test and verify the effectiveness of seismic retrofitting measures after installation

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. seismic zones of India as per IS 1893 and earthquake damage types such as soft storey failure and short column effect

KU2. importance of seismic retrofitting in reducing the vulnerability of existing buildings to earthquake damage

KU3. seismic vulnerability assessment parameters including building age, construction type, structural system, and geographic seismic zone classification

KU4. types of seismic retrofitting systems including shear walls, moment-resisting frames, bracing systems (X-bracing, K-bracing, chevron bracing), and base isolation systems

KU5. components and structural function of reinforced concrete and masonry shear wall systems and their installation requirements

KU6. types of structural steel bracing systems for seismic resistance including X-bracing, K-bracing, and chevron bracing, and their installation requirements



Qualification Pack

- KU7.** principles of base isolation systems including types of isolators (lead-rubber bearings, friction pendulum systems) and their installation requirements
- KU8.** relevant IS codes for seismic design and retrofitting including IS 1893 and IS 13920
- KU9.** testing and verification methods used to assess the effectiveness and quality of seismic retrofitting installations
- KU10.** visual inspection criteria for verifying installed seismic retrofitting measures including checks for correct placement and dimensions
- KU11.** workmanship checks for fastening quality and surface finish of installed seismic retrofitting measures as per engineering drawings
- KU12.** the interaction and compatibility of seismic retrofitting measures with the existing structural system of the building

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and follow engineering drawings, work instructions, and site safety requirements
- GS2.** write accurate records of work activities, observations, and outcomes
- GS3.** communicate work progress and non-conformances to supervisors promptly
- GS4.** listen to and follow instructions from supervisors and engineers
- GS5.** identify safety hazards particularly when working at height and report to supervisors
- GS6.** plan and sequence work activities to minimise disruption to building occupants
- GS7.** work cooperatively with multiple trades and team members on site
- GS8.** check completed work against drawings and report any discrepancies to supervisors
- GS9.** use inspection tools and measurement instruments to check installed work against specifications
- GS10.** adhere strictly to seismic safety codes and site instructions during all installation activities



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Analyse seismic vulnerability of existing buildings</i>	8	15	-	2
PC1. analyze existing buildings for seismic vulnerability based on building age, design, and location	-	-	-	-
<i>Install, ensure compliance and verify seismic retrofitting measures</i>	22	45	-	8
PC2. install seismic retrofitting measures such as shear walls, base isolation systems, or bracing systems as required by the building's seismic evaluation	-	-	-	-
PC3. ensure that retrofitting techniques comply with seismic safety codes and standards	-	-	-	-
PC4. test and verify the effectiveness of seismic retrofitting measures after installation	-	-	-	-
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CBWE/CON/N0105
NOS Name	Implement Seismic Retrofitting Solutions
Sector	Construction
Sub-Sector	
Occupation	Building Retrofitting
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

CBWE/CON/N0106: Execute Quality Control and Safety Standards in Retrofitting Work

Description

This unit describes the skills and knowledge required to conduct quality control checks during and after repair and retrofitting processes, maintain accurate records of materials and methods, adhere to safety protocols including use of personal protective equipment, and ensure that retrofitting work is carried out in compliance with environmental guidelines.

Scope

The scope covers the following :

- Conduct quality control checks and maintain records
- Adhere to safety protocols and environmental standards

Elements and Performance Criteria

Conduct quality control checks and maintain records

To be competent, the user/individual on the job must be able to:

- PC1.** conduct quality control checks during and after the repair and retrofitting processes to ensure compliance with project specifications
- PC2.** maintain accurate records of materials used, methods applied, and progress in the retrofitting work

Adhere to safety protocols and environmental standards

To be competent, the user/individual on the job must be able to:

- PC3.** adhere to safety protocols by using personal protective equipment (PPE) and ensuring that the worksite is free of hazards
- PC4.** ensure that retrofitting work is carried out in compliance with environmental guidelines

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** quality control principles, methods, and inspection stages applicable to construction and retrofitting work
- KU2.** types of quality control checks for retrofitting work including visual inspection, dimensional verification, bond pull-off tests, and material test certificates
- KU3.** record-keeping requirements for retrofitting projects including daily progress reports, material registers, test reports, and non-conformance reports
- KU4.** types of PPE required for retrofitting work including helmets, safety harnesses, gloves, goggles, dust masks, and ear protection
- KU5.** relevant safety legislation for construction and retrofitting work including the BOCW (Regulation of Employment and Conditions of Service) Act 1996



Qualification Pack

- KU6.** environmental guidelines for retrofitting work including waste segregation, hazardous material handling, and dust and noise control
- KU7.** purpose of toolbox talks and safety briefings before commencing work on a retrofitting site
- KU8.** topics to be covered in a daily toolbox talk including hazards, PPE, work sequence, and emergency procedures
- KU9.** emergency response actions for common site accidents including falls, chemical exposure, cuts, and structural incidents
- KU10.** role and responsibilities of the retrofitting technician during a site emergency
- KU11.** site housekeeping standards for retrofitting worksites including material storage, stacking requirements, and passage clearance
- KU12.** waste segregation and disposal procedures applicable to retrofitting sites as per BOCW Act provisions
- KU13.** non-conformance identification criteria, documentation format, and reporting obligations under site quality procedures

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and follow quality checklists, safety regulations, and work instructions
- GS2.** write accurate and complete records, reports, and site documentation
- GS3.** communicate safety observations and quality concerns to supervisors clearly
- GS4.** listen to and follow instructions from supervisors
- GS5.** identify safety hazards and environmental risks and report to supervisors immediately
- GS6.** plan and schedule work activities to cover all critical stages within the agreed time
- GS7.** complete work as per agreed quality standards and within the required time frame
- GS8.** work cooperatively with team members and encourage safe working practices on site
- GS9.** conduct checks at defined hold points during work and record outcomes accurately
- GS10.** segregate and dispose of waste materials as per site environmental procedures



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conduct quality control checks and maintain records</i>	15	30	-	5
PC1. conduct quality control checks during and after the repair and retrofitting processes to ensure compliance with project specifications	-	-	-	-
PC2. maintain accurate records of materials used, methods applied, and progress in the retrofitting work	-	-	-	-
<i>Adhere to safety protocols and environmental standards</i>	15	30	-	5
PC3. adhere to safety protocols by using personal protective equipment (PPE) and ensuring that the worksite is free of hazards	-	-	-	-
PC4. ensure that retrofitting work is carried out in compliance with environmental guidelines	-	-	-	-
NOS Total	30	60	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CBWE/CON/N0106
NOS Name	Execute Quality Control and Safety Standards in Retrofitting Work
Sector	Construction
Sub-Sector	
Occupation	Building Retrofitting
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for this Qualification Pack are laid down by Dattopant Thengadi National Board for Workers Education and Development (DTNBWED) as the Awarding Body. Each Performance Criteria (PC)/Element is assigned marks proportional to its importance in the NOS. The Awarding Body shall lay down the proportion of marks for Theory and Practical for each Element/PC.
2. The assessment for the knowledge part shall be based on a question bank created by Assessment Agencies and approved by the Awarding Body.
3. Individual assessment agencies shall create unique question papers for the knowledge/theory part for assessment of candidates as per the assessment criteria specified below.
4. Individual assessment agencies shall create unique evaluations for skill practical for every candidate at each examination/training centre based on the assessment criteria.
5. To pass the Qualification Pack assessment, every trainee should score the recommended aggregate for the QP.
6. The Assessor shall check the final outcome of the practical performance while evaluating the steps



Qualification Pack

performed to achieve the final outcome.

7. The trainee shall be provided with a chance to repeat the test to correct procedures in case of improper performance.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CBWE/CON/N0101.Assess Building Condition and Identify Repair & Strengthening Requirements	30	60	-	10	100	15
CBWE/CON/N0102.Conduct Structural Repairs (Masonry, Plastering, Waterproofing, etc.)	30	60	-	10	100	20
CBWE/CON/N0103.Perform Crack and Joint Repairs (Epoxy, Polyurethane Injection)	30	60	-	10	100	15
CBWE/CON/N0104.Apply Structural Re-Strengthening Techniques (Steel, Concrete, FRP)	30	60	-	10	100	20
CBWE/CON/N0105.Implement Seismic Retrofitting Solutions	30	60	-	10	100	20
CBWE/CON/N0106.Execute Quality Control and Safety Standards in Retrofitting Work	30	60	-	10	100	5
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	200	390	-	60	650	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
FRP	Fiber-Reinforced Polymer
GFRP	Glass Fiber Reinforced Polymer
IS	Indian Standard
PPE	Personal Protective Equipment



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.